Auditing a physics-informed augmentation claim: from +9.8% to a diagnosed negative result

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Abstract

A thermal solar-module classifier reported a +9.8 percentage-point accuracy gain from heat-equation–based synthetic augmentation over a baseline. Under rigorous evaluation the claim does not survive. Roughly +9 points are a training-schedule artifact (an undertrained baseline); the residual ~ 1.9 points are a *rebalancing* effect that plain image duplication matches or beats, and they do not improve recall on the rare fault classes the method targets. Two diagnostics localise the cause: the synthetic images are trivially separable from real ones (discriminator AUC 0.97), and class weighting already handles imbalance more effectively than synthetic rebalancing. A realism upgrade to the generator, although visually more convincing, measured *worse* (AUC 1.00). We conclude with the implied redesign: let physics supply fault *structure* and let a network learn the *appearance*.

1 Setup and metrics

- **Data.** InfraredSolarModules [1]: $\sim$20,000 24×40 thermal IR crops, 12 classes, heavily imbalanced ($\sim$50% `No-Anomaly`; four rare fault classes have only $\sim$120–170 training images each).
- **Model.** EfficientNet-B0 [2] fine-tuned from ImageNet; inverse-frequency class-weighted cross-entropy; warmup $\rightarrow$ cosine schedule; stratified 70/15/15 splits.
- **Harness.** All comparisons are paired on one fixed test set, run over 3 seeds, with bootstrap 95% confidence intervals on *per-class recall*.
- **Key framing.** The project's objective is *rare-fault detection*, so the right metric is **rare-class recall**, not accuracy (which is dominated by the majority class). This distinction drives every conclusion below.

2 The headline is mostly a schedule artifact

The original baseline is undertrained: a slow cosine schedule plus patience-5 early stopping halts it at epoch 12. Fixing only the schedule (a faster 15-epoch cosine) recovers almost the entire gap with *no* synthetic data (Table 1).

Run	Schedule	Test acc.
baseline (as reported)	30 ep	0.690
augmented (as reported)	30 ep	0.788
clean, retuned schedule	15 ep	0.784
augmented, retuned schedule	15 ep	0.808

Table 1: The +9.8-point headline is mostly the difference between an undertrained and a well-trained baseline. A clean model on a 15-epoch schedule already reaches 0.784.

3 The backbone choice holds up

EfficientNet-B0 is not assumed: it wins a sweep over four `timm` backbones on identical clean data, *and* is the smallest. Larger backbones overfit this small, imbalanced set (Table 2).

Backbone	Params (M)	Test acc.	Macro F1
EfficientNet-B0	**4.0**	**0.784**	**0.658**
MobileNetV3-Large	4.2	0.760	0.646
ConvNeXt-Tiny	27.8	0.745	0.635
ResNet-50	23.5	0.694	0.577

Table 2: Backbone sweep (clean data, 15-epoch schedule). Smallest model wins.

4 Honest baselines: physics is no better than oversampling

The decisive test compares physics augmentation against the *cheap* ways to rebalance. `oversample` duplicates real rare-class images to the same target count as the synthesiser, isolating the synthetic *signal* from mere rebalancing; `randaugment` is a strong generic-augmentation control. We also toggle class weighting off ("no-weights") to probe its interaction with rebalancing (Table 3).

Condition	Weights	Acc.	Macro F1	Macro rec.	**Rare rec.**
clean	yes	0.782	0.658	0.699	**0.612**
oversample	yes	0.807	0.678	0.697	0.574
randaugment	yes	0.782	0.654	0.702	0.615
physics	yes	0.800	0.668	0.693	0.569
clean	no	0.839	0.699	0.684	0.560
oversample	no	0.826	0.685	0.680	0.582
physics	no	0.837	0.700	0.686	0.568

Table 3: Honest baselines, 15-epoch schedule, 3-seed means. Physics is statistically indistinguishable from `oversample` on every metric, and is the *worst* condition on rare-class recall.

Findings:

- **Physics does not beat duplication.** On accuracy, `oversample` (0.807) edges out `physics` (0.800); elsewhere they are within noise. The synthetic pixels add nothing over copy-paste.
- **The accuracy gain is a rebalancing artifact, not rare-fault improvement.** Physics' +1.9 points over `clean` (3-seed paired $+0.019 \pm 0.006$) come from the *majority* class (`No-Anomaly` recall $0.85 \rightarrow 0.88$); on rare-class recall physics is worst, and significantly below `clean` (paired $\Delta = +0.043$, 95% CI $[0.003, 0.082]$). See Fig. 1.
- **Class weighting is the strongest single lever for rare classes.** `clean` with weights (0.612) beats every other condition. Turning weights off trades rare recall for accuracy ($0.78 \rightarrow 0.84$). Data rebalancing is a weaker *substitute* that *hurts* when stacked on weighting — exactly the current configuration.

5 Diagnostics: why it fails

Two cheap diagnostics localise the failure to *two independent* causes.

D1 — domain gap (Fig. 2). A classifier separates synthetic from real images of the same classes at AUC 0.966. The gap survives heavy low-pass blurring (0.934 at radius 2, 0.867 at radius 4), so it lives in the *coarse shape* of the steady-state heat blobs, not in seams or compression: the synthetic faults simply do not look like real faults.

D2 — redundant rebalancing. The "no-weights" rows of Table 3 confirm that class weighting and data rebalancing are *substitutes* for handling imbalance; stacking them over-corrects toward the majority and lowers rare recall. Even a perfect synthesiser would have to be deployed without fighting the class weights.

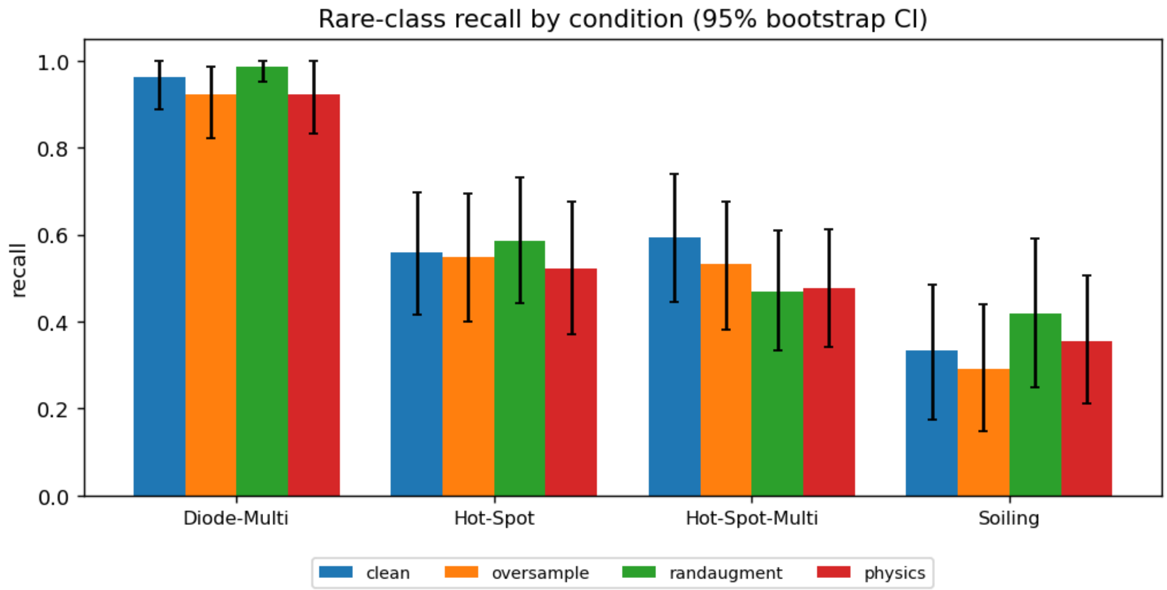

Figure 1: Rare-class recall with 95% bootstrap CIs. Physics (red) is never the best bar. Wide intervals (each rare class has ~ 25 test images) are themselves a finding: the evaluation is underpowered, which is why a metric-appropriate, CI-based harness was necessary.

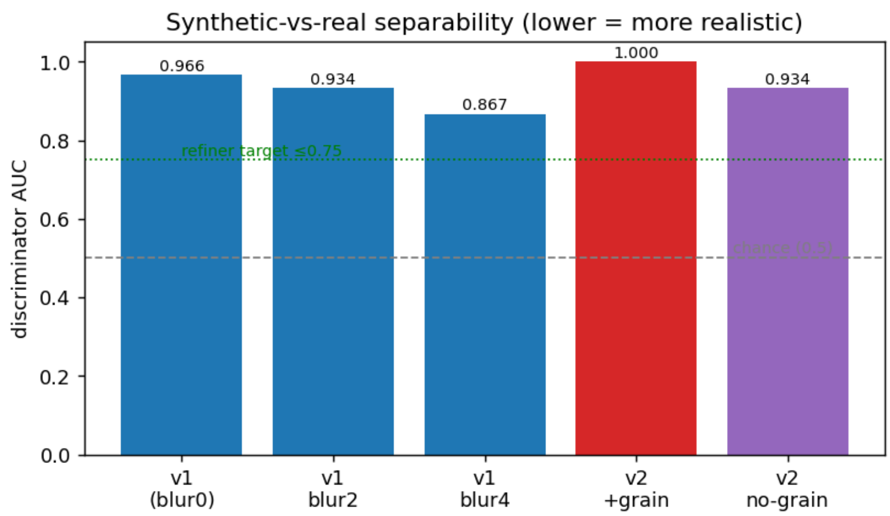

Figure 2: Synthetic-vs-real discriminator AUC (lower is more realistic). The v2 realism upgrade made the gap *worse*.

6 A realism upgrade that measured worse (v2)

A second generator (`heat_equation_v2`) replaced steady-state blobs with *transient* diffusion, fixed an aspect-ratio warp, calibrated per-class contrast to real data, and corrected the class models (real "Soiling" is bright and streaky, not a smooth darkening). It looks markedly more realistic (Fig. 3) — yet the discriminator scored it *worse*: AUC 1.000 with the (naïvely added, i.i.d.) sensor grain, and 0.934 once the grain was ablated. Lessons:

- **Visual realism is misleading;** the objective discriminator is the right judge. Adding i.i.d. grain on top of a base that already carries real, spatially-correlated noise is an obvious tell.
- **The additive-blend-on-clean-base paradigm is near a detectability ceiling:** corrected, v2 (0.934) is no better than v1 (0.966). Incremental geometry tweaks will not close the gap.

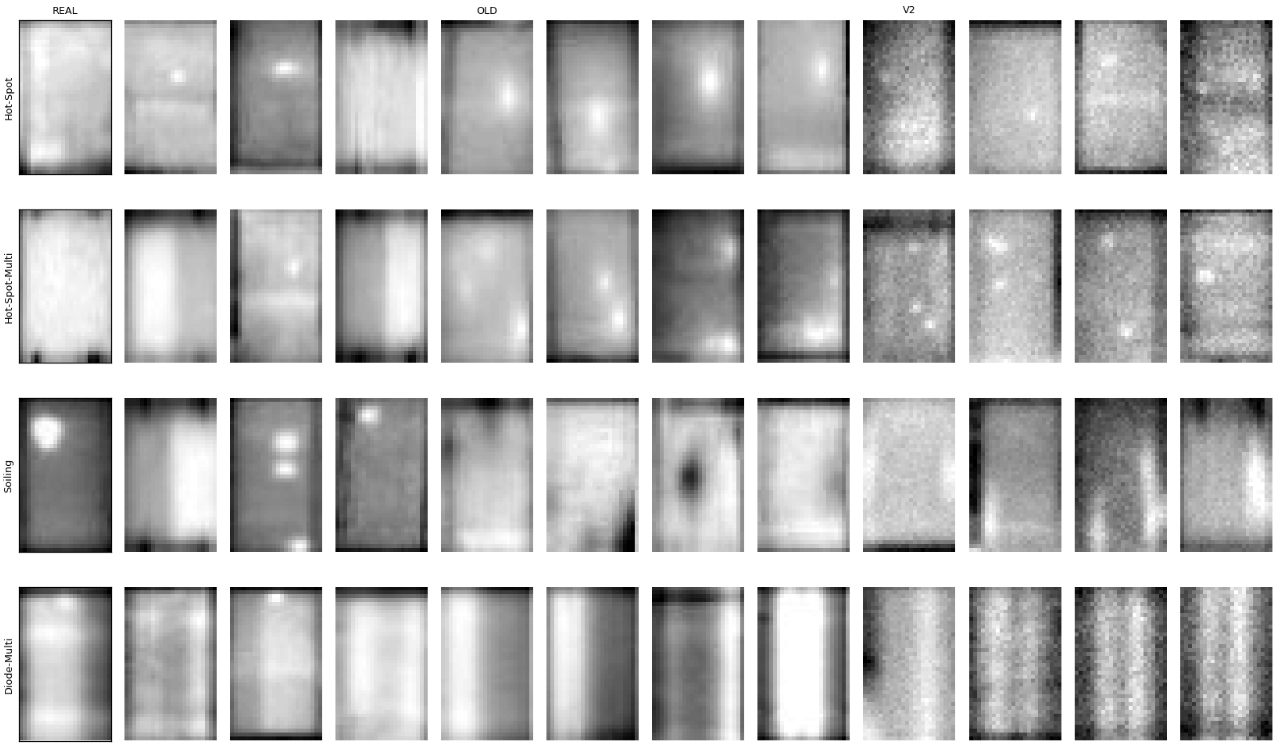

Figure 3: Real | v1-synthetic | v2-synthetic for the four augmented classes. v2 is visually closer, but measured more separable — the cautionary core of this report.

7 Discussion: the right role for physics

The error was asking physics to generate *pixels*. A hand-specified PDE is good at *structure* (where heat is, how it diffuses — which carries the label) and bad at *appearance* (sensor texture, noise statistics, tone mapping). The defensible division of labour is therefore: **physics proposes the fault structure; a learned model renders the appearance.** This keeps physics genuinely load-bearing rather than decorative, and turns the realism gap into a learning target instead of a hand-tuning chore.

8 Future work: a learned refiner (SimGAN)

We propose a SimGAN-style refiner [3]: a small fully-convolutional residual network R takes a physics-generated image and outputs a more realistic one, trained adversarially against a patch discriminator *plus* a self-regularization term $\lambda\|R(x) - x\|_1$ that preserves the physics-determined fault (hence the label). The discriminator already built for D1 doubles as both the adversary and — crucially, with an *independently trained* copy — the score-card.

Validation is staged and honestly gated:

1. **Realism gate.** Refine the synthetic set and re-score with a fresh discriminator. Success: AUC falls from 0.93 toward ≤ 0.75.

2. **Utility gate** (only if the first passes). Add a `physics_refined` condition to the harness and test rare-class recall against `oversample`, with weights off (per D2). Success: a positive paired delta whose CI excludes zero.

Because the heat solver is a stack of Laplacian convolutions — a fixed-weight CNN — the entire pipeline is differentiable, enabling a later variant that learns the source field or PDE parameters end-to-end through the physics. The binding constraints remain: ~ 600 real fault images is a tight data budget, and any generator must clear *both* gates, not just the first. If the realism gate fails, the negative result documented here stands — which is itself an acceptable, well-characterised outcome.

References

[1] RaptorMaps, *InfraredSolarModules: a dataset for infrared solar module anomaly classification*, `https://github.com/RaptorMaps/InfraredSolarModules`, 2020.

[2] M. Tan and Q. V. Le, "EfficientNet: rethinking model scaling for convolutional neural networks", in International conference on machine learning (icml) (2019), arXiv:1905.11946.

[3] A. Shrivastava, T. Pfister, O. Tuzel, J. Susskind, W. Wang, and R. Webb, "Learning from simulated and unsupervised images through adversarial training", in Ieee conference on computer vision and pattern recognition (cvpr) (2017), arXiv:1612.07828.